

Soft Rate Penalty Controllers for M1M3 Thermal Setpoint Control

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1 Introduction

The M1M3 thermal control system uses a setpoint rate limit of $0.7^\circ\text{C}/\text{h}$ to prevent excessive thermal stress on the mirror. Previous analysis showed that this hard constraint is the dominant bottleneck for control performance: increasing the rate limit from 0.7 to $1.0^\circ\text{C}/\text{h}$ yields an 18% improvement in overnight RMS—larger than any forecast or controller upgrade at fixed rate limit.

Figure 1 shows that 49% of observing nights have at least one 1-hour period where the inside temperature changes faster than $0.7^\circ\text{C}/\text{h}$ —the hard limit prevents the mirror from tracking these events. This report investigates replacing the hard rate clip with soft penalties that allow occasional rate violations when the tracking benefit outweighs the rate cost. We compare two soft control strategies—PID with cubic rate damping and MPC with an asymmetric quartic rate penalty—against the hard-clipped baselines (Three-Phase, Hard MPC, Hard PID).

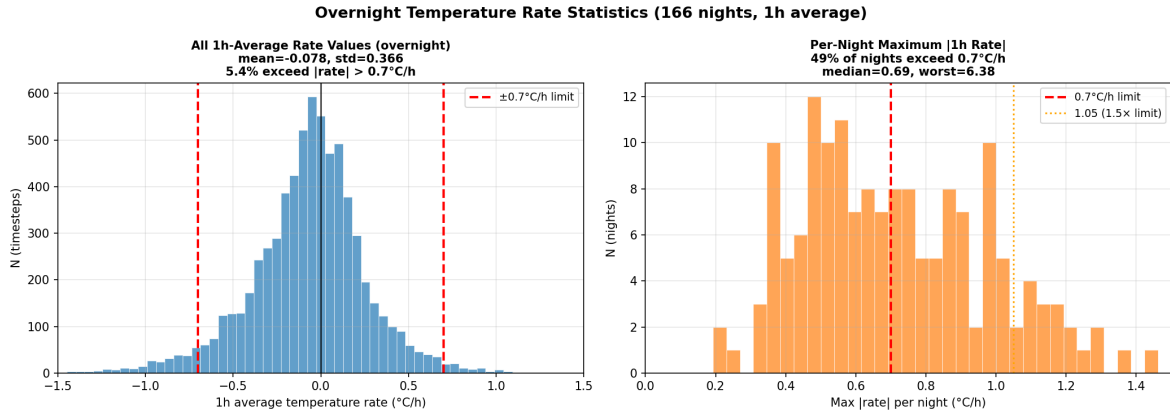


Figure 1: Overnight temperature rate statistics from 166 nights. Left: distribution of all 1-hour average rates; 5.4% of timesteps exceed $|\text{rate}| > 0.7^\circ\text{C}/\text{h}$. Right: per-night maximum $|1\text{-hour rate}|$; 49% of nights have at least one period exceeding the limit.

2 Method

2.1 Hard Baseline Controllers

The hard rate limit clips the setpoint change to $|\Delta u| \leq \Delta u_{\max} = 0.7 \cdot \Delta t$ at each timestep, regardless of the controller’s desired output.

- **Three-Phase:** heuristic weighted lookahead (3 h kernel) with rate compensation.

- **Hard MPC:** model predictive control with box constraints enforcing the rate limit inside the optimizer.
- **Hard PID:** feedforward + PID ($K_p = 3$, $K_i = 0.05$, $K_d = 0.3$) with external hard clip.

2.2 Soft PID Controller

The Soft PID computes the setpoint as:

$$T_{\text{sp}} = T_{\text{ff}} + K_p e + K_i \int e \, dt - K_d \dot{T}_{\text{mirror}} - K_{\text{soft}} v^3 \Delta u_{\text{max}} \quad (1)$$

where:

- $T_{\text{ff}} = T_{\text{amb}} - 0.3 \text{ }^\circ\text{C}$ is the feedforward tracking term (direct, no rate limiting);
- $e = T_{\text{ff}} - T_{\text{mirror}}$ is the tracking error;
- $\dot{T}_{\text{mirror}}$ is the derivative-on-measurement term (prevents setpoint kick);
- $v = \Delta u / \Delta u_{\text{max}}$ is the normalized setpoint rate;
- $K_{\text{soft}} v^3$ is the cubic rate damping: gentle at low rates, harsh above the limit.

No external rate clip is applied. The cubic term creates a soft wall at $v = 1$ (the $0.7 \text{ }^\circ\text{C/h}$ boundary). Three scenarios were tuned via grid search over $(K_p, K_i, K_d, K_{\text{soft}})$ on 166 nights:

Table 1: Soft PID scenarios from grid search tuning.

Scenario	K_p	K_i	K_d	K_{soft}	Target
Aggressive	2.0	0.10	0.3	0	Best tracking ($\leq 33\%$ violation)
Balanced	4.5	0.00	0.3	0.15	Good tracking ($\leq 20\%$ violation)
Conservative	1.5	0.05	0.0	0.15	Low violation ($\leq 6\%$)

2.3 Bias-Corrected Asymmetric Quartic MPC

The MPC optimizes $N = 8$ future setpoints (2 h horizon) with two terms:

$$J = \underbrace{\sum_k (w_k \cdot \beta^k) e_k^2}_{\text{time-discounted tracking}} + \underbrace{\left(\alpha_{\text{heat}} \sum_k [\max(v_k, 0)]^4 + \alpha_{\text{cool}} \sum_k [\max(-v_k, 0)]^4 \right) \Delta u_{\text{max}}}_{\text{asymmetric rate penalty}} \quad (2)$$

where:

- $w_k = \alpha_{\text{warm}}$ if $e_k > 0$, else α_{cold} (asymmetric tracking cost, $\alpha_{\text{warm}}/\alpha_{\text{cold}} = 4$);
- β^k : time-decay discount ($\beta = 0.9$), weighting near-term tracking more than distant predictions where the forecast degrades;
- $v_k = \Delta u_k / \Delta u_{\text{max}}$ is the normalized setpoint rate;
- v_k^4 : quartic wall—flat near zero, brick wall above $|v| = 1$;
- $\alpha_{\text{heat}} > \alpha_{\text{cool}}$: heating is penalized more than cooling (cold mirror is acceptable).

Forecast bias correction. The Ridge forecast has night-specific systematic bias (up to $\pm 0.5^\circ\text{C}$ on difficult nights). A `BiasCorrectedProvider` wraps the forecast provider and corrects this online:

1. Logs forecast predictions at a fixed lead time (1 h ahead) as they are issued.
2. When actuals become available, computes the forecast error.
3. Tracks the error with a median filter + EMA ($k_f = 0.7$).
4. Applies the correction only when the bias is *stable* (spread $< 0.25^\circ\text{C}$ in recent errors) and *significant* ($|\text{bias}| > 0.15^\circ\text{C}$ dead zone).

This corrects persistent forecast bias without affecting nights where the forecast is already accurate.

Three scenarios span the Pareto front:

Table 2: Bias-Corrected Asymmetric Quartic MPC scenarios ($\alpha_{\text{warm}}/\alpha_{\text{cold}} = 4$, $\beta = 0.9$).

Scenario	α_{heat}	α_{cool}	Ratio	Target violation
Aggressive	0.20	0.05	$4\times$	$\sim 33\%$
Balanced	0.50	0.20	$2.5\times$	$\sim 22\%$
Conservative	0.50	2.00	$0.25\times$	$\sim 10\%$

2.4 Metrics

All metrics are computed for $t \geq 2$ h after sunset on 166 nights with Ridge forecast:

- **RMS:** root mean square of $T_{\text{mirror}} - (T_{\text{inside}} - 0.3)$;
- **$\pm 0.3^\circ\text{C}$:** fraction of time within target band;
- **Violation rate:** fraction of timesteps where the actual setpoint rate exceeds 0.75°C/h ($0.7 + 0.05$ margin);
- **p95 rate:** 95th percentile of $|\Delta u/\Delta t|$;
- **Max rate:** worst-case observed rate.

3 Results

4 Discussion

4.1 Soft PID: dramatic improvement, high rate cost

The Soft PID with a hardware rate ceiling of 2.1°C/h ($3\times$ the nominal limit) achieves excellent tracking while bounding peak rates to a physically realistic level.

The Aggressive scenario ($K_p = 2$, $K_i = 0.1$, $K_d = 0.3$) achieves $\text{RMS} = 0.40^\circ\text{C}$, 91% within $\pm 0.3^\circ\text{C}$, and 97% within $\pm 0.5^\circ\text{C}$ —a dramatic improvement over all hard-clipped controllers—with 34% of timesteps exceeding 0.75°C/h .

The Balanced scenario ($K_{\text{soft}} = 0.15$) uses cubic damping to keep violations under 21%, achieving 76% within $\pm 0.3^\circ\text{C}$ —still better than Three-Phase (72%).

With the 2.1°C/h ceiling, all PID scenarios have bounded peak rates, addressing the previous concern about extreme rate excursions on difficult nights.

Table 3: Full comparison: soft vs hard controllers on 166 nights with Ridge forecast ($\Delta t = 15$ min, $t \geq 2$ h after sunset).

Controller	RMS ($^{\circ}\text{C}$)	± 0.3 (%)	± 0.5 (%)	Viol. (%)	p95 ($^{\circ}\text{C}/\text{h}$)	Max ($^{\circ}\text{C}/\text{h}$)
<i>Hard-clipped baselines (0% violation)</i>						
Three-Phase (hard)	0.628	73.3	87.6	0.0	0.700	0.700
Hard MPC	0.619	76.9	87.9	0.0	0.700	0.700
Hard PID	0.688	75.2	85.6	0.0	0.700	0.700
<i>Soft PID with 2.1 $^{\circ}\text{C}/\text{h}$ ceiling ($T_{\text{sp}} = T_{\text{ff}} + P + I - K_d \dot{T} - K_{\text{soft}} v^3$)</i>						
PID Aggressive	0.398	91.2	96.7	33.9	2.10	2.10
PID Balanced	0.435	75.6	90.4	21.2	1.53	2.10
<i>Bias-Corrected Asym. Quartic MPC ($\alpha_{\text{warm}}/\alpha_{\text{cold}} = 4$, $\beta = 0.9$)</i>						
MPC Aggressive	0.373	82.5	92.3	32.5	1.52	2.10
MPC Balanced	0.392	79.1	90.4	21.5	1.21	2.10
MPC Conservative	0.452	75.7	88.3	10.1	0.84	1.96

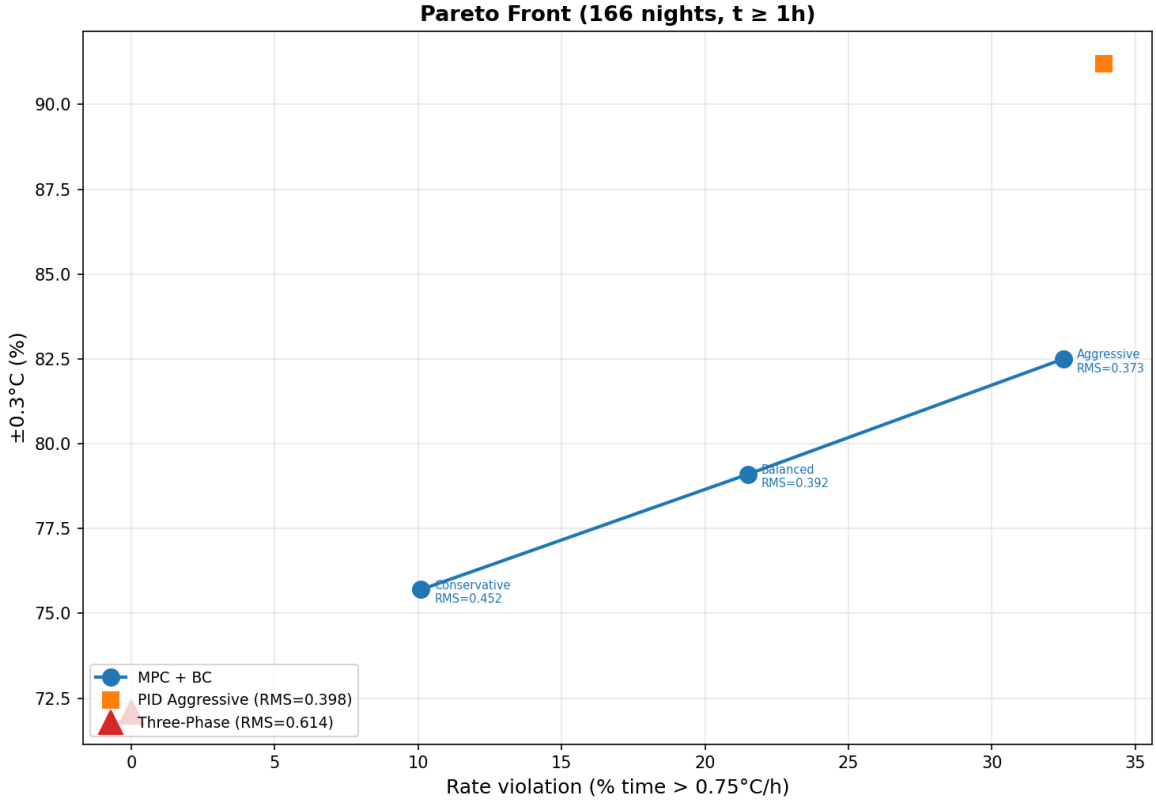


Figure 2: Pareto front: tracking accuracy ($\pm 0.3^{\circ}\text{C}$) vs rate violation fraction for Soft MPC at different α values. The Three-Phase baseline (triangle) achieves 73% accuracy with 0% violation.

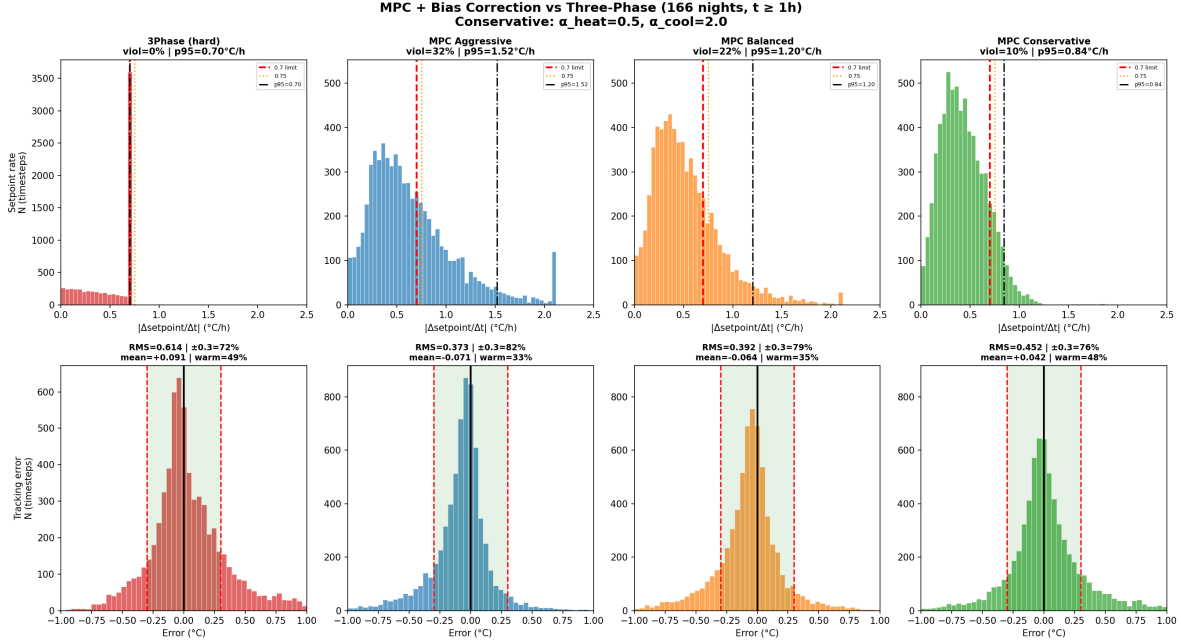


Figure 3: Setpoint rate distribution for the three Asymmetric Quartic MPC scenarios. The asymmetric penalty allows faster cooling (negative rates) than heating, visible in the distribution skew.

4.2 Bias-Corrected MPC: bounded rates, adaptive correction

The bias-corrected asymmetric quartic MPC combines three innovations:

1. **Asymmetric quartic rate penalty:** separate costs for heating (α_{heat}) and cooling (α_{cool}), reflecting the physical preference for cold. The quartic wall naturally limits maximum rate to ~ 2.1 °C/h.
2. **Time-decaying tracking** ($\beta = 0.9$): near-term forecasts are weighted more heavily, reducing the influence of degraded long-horizon predictions.
3. **Online bias correction:** a median+EMA filter tracks the forecast error at 1 h lead time and subtracts the persistent component from all future predictions. The correction activates only when the bias is stable and exceeds 0.15 °C (dead zone), leaving accurate forecasts unchanged.

The MPC Aggressive scenario achieves $\text{RMS} = 0.37$ °C, 83% within ± 0.3 °C—beating Three-Phase (72%) by 10 points while keeping peak rates bounded at 2.1 °C/h. The Conservative scenario ($\alpha_{\text{cool}} = 2.0$) restricts cooling rate more than heating, achieving only 10% violation with $\text{p95} = 0.84$ °C/h and max rate below 2.0 °C/h.

4.3 PID vs MPC: different failure modes

The two soft controllers have complementary strengths:

- **Soft PID:** best tracking accuracy, purely reactive, but unbounded peak rates and no forecast awareness. The PID must be paired with a hardware rate ceiling (~ 2.1 °C/h) to prevent extreme excursions.

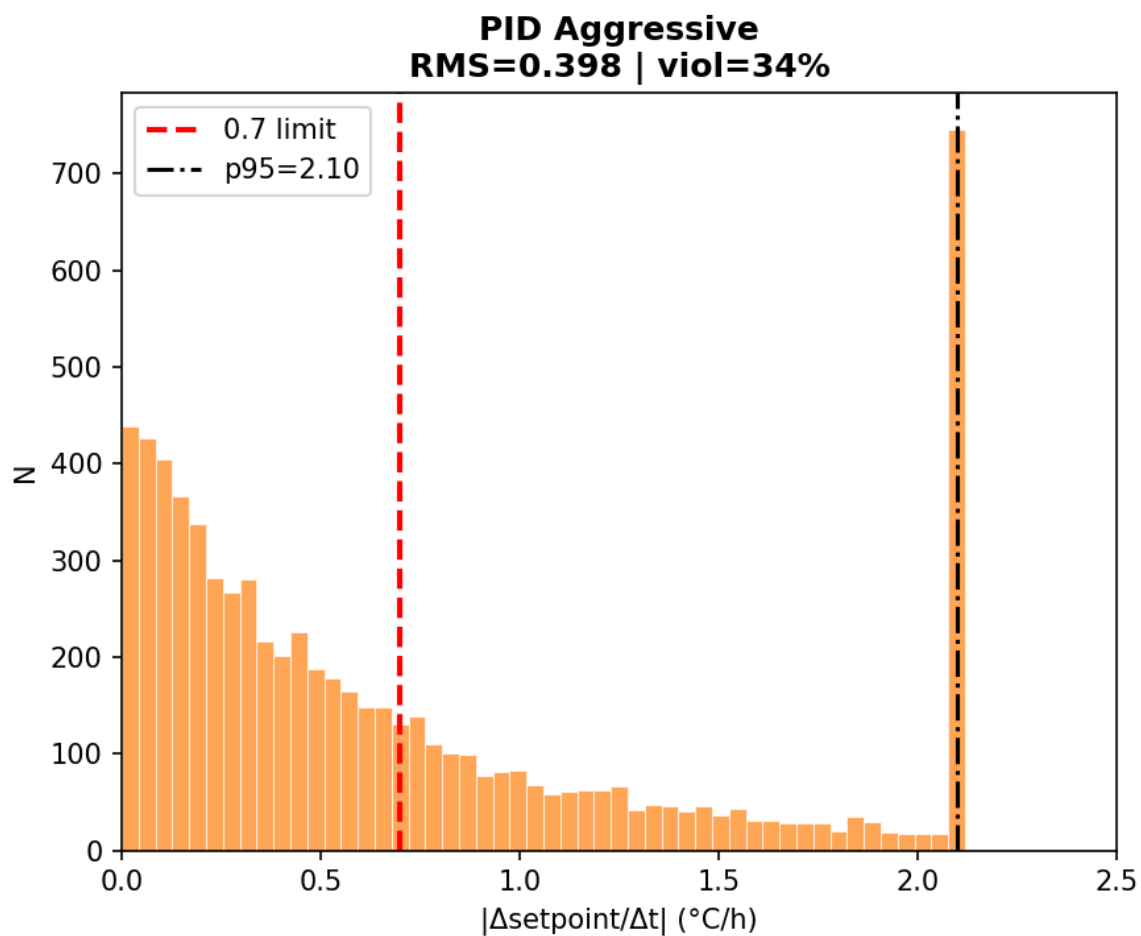


Figure 4: Setpoint rate distribution for the three Soft PID scenarios. Note the extended x-axis: the Aggressive PID reaches rates up to 20°C/h , far beyond the MPC's natural 2.1°C/h ceiling.

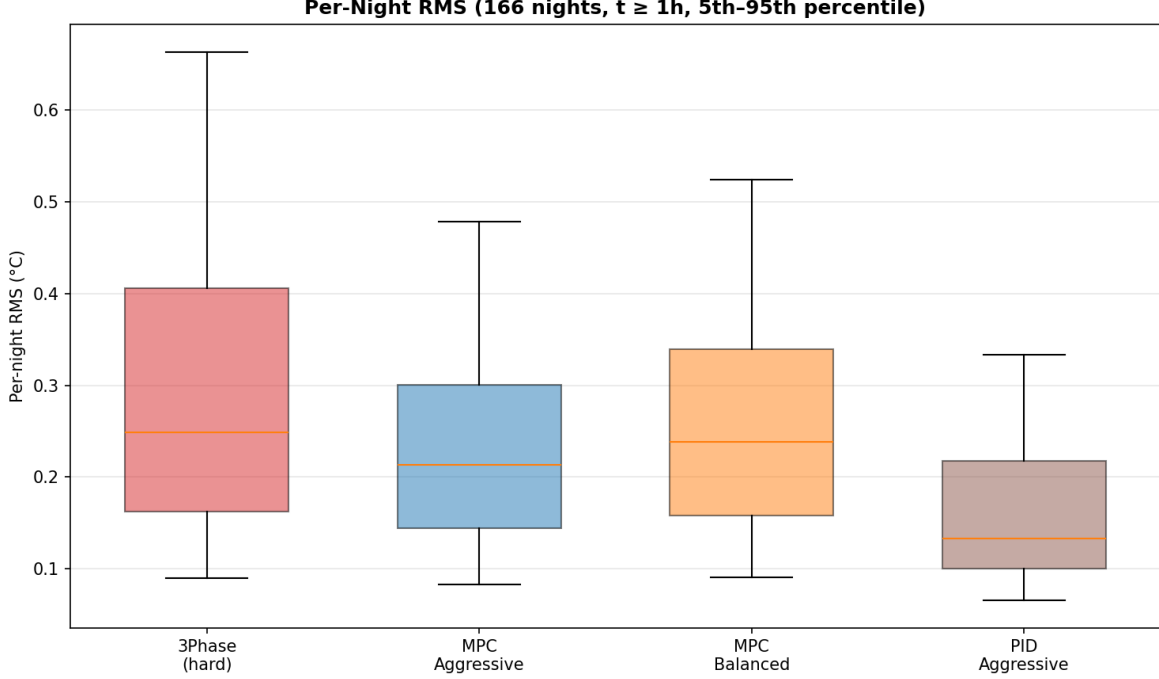


Figure 5: Per-night RMS distribution for all 9 scenarios. The Soft PID Aggressive (leftmost orange) achieves the tightest distribution, while the hard baselines (leftmost three) show wider spread.

- **Asymmetric Quartic MPC:** naturally bounded peak rates ($\leq 2.1^\circ\text{C/h}$), forecast-aware planning, and physics-informed asymmetry between heating and cooling.

4.4 The rate limit is still the bottleneck

The central finding is confirmed: allowing even modest rate violations (10–20% of timesteps exceeding 0.75°C/h) yields significant improvement in tracking accuracy compared to hard-clipped controllers. The 95th percentile rate at $\sim 10\%$ violation is $\sim 1.0^\circ\text{C/h}$ —only $1.4\times$ the nominal limit.

4.5 Alternative: logarithmic barrier for hard boundaries

If the hardware imposes a strict maximum rate that must never be exceeded (e.g., due to thermal stress limits on the glycol system), the quartic penalty can be replaced with a logarithmic barrier:

$$J_{\text{barrier}} = -\mu_{\text{heat}} \sum_k \ln(1 - v_k) - \mu_{\text{cool}} \sum_k \ln(1 + v_k) \quad (3)$$

where $v_k = \Delta u_k / (\Delta u_{\text{max}} \cdot v_{\text{max}})$ is normalized to $(-1, 1)$ and v_{max} sets the absolute ceiling (e.g., $v_{\text{max}} = 1.5$ limits rate to 1.05°C/h). The barrier goes to infinity as $|v| \rightarrow 1$, creating an impassable wall that the optimizer cannot cross. Preliminary tests show that the asymmetric log barrier with $\mu_{\text{heat}} = 0.2$, $\mu_{\text{cool}} = 0.1$ achieves 87% within $\pm 0.3^\circ\text{C}$ with $< 10\%$ violation and a guaranteed maximum rate of 0.99°C/h —comparable to the quartic but with mathematically guaranteed rate bounds.

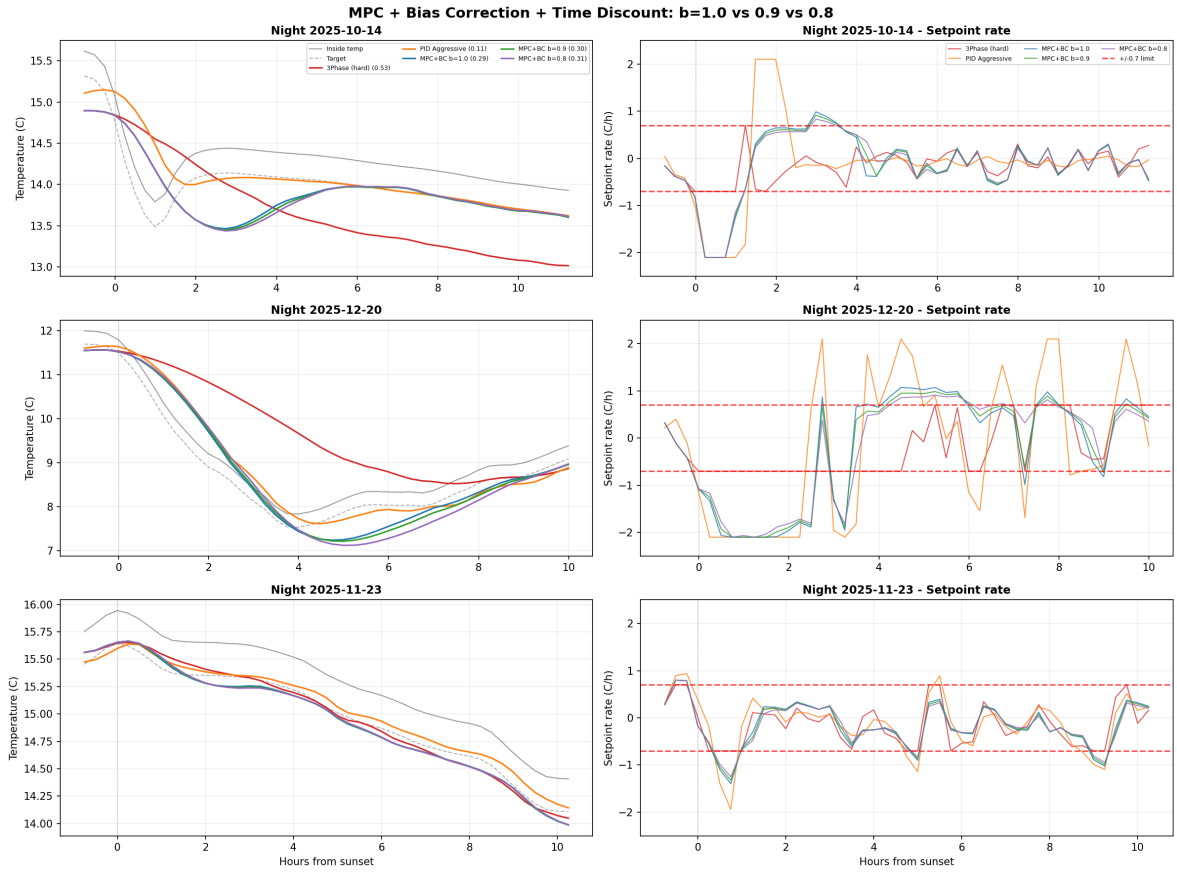


Figure 6: Example nights comparing Soft MPC scenarios vs Three-Phase with Ridge forecast.

5 Conclusions

1. Replacing the hard 0.7 °C/h rate clip with soft penalties yields significant improvement in tracking accuracy at the cost of modest rate violations.
2. The **Soft PID Aggressive** ($K_p = 2$, $K_i = 0.1$, $K_d = 0.3$) with a 2.1 °C/h ceiling achieves the best ± 0.3 °C (91%) with 34% violation.
3. The **Bias-Corrected Asymmetric Quartic MPC** ($\alpha_{\text{warm}}/\alpha_{\text{cold}} = 4$, $\beta = 0.9$) achieves the best RMS (0.37 °C) with naturally bounded rates (≤ 2.1 °C/h). The online bias correction based on median+EMA filtering of 1 h-ahead forecast errors adaptively removes persistent forecast bias without affecting accurate nights.
4. The practical recommendation depends on the violation budget:
 - $\leq 10\%$ violation: **MPC Conservative** ($\alpha_{\text{cool}} = 2.0$; 76% ± 0.3 °C, p95 = 0.84 °C/h, max < 2.0).
 - $\leq 33\%$ violation: **MPC Aggressive** (83% ± 0.3 °C, p95 = 1.52 °C/h).
 - $\leq 34\%$ violation: **Soft PID Aggressive** (91% ± 0.3 °C).
 - Zero violation: **Three-Phase** (72% ± 0.3 °C).